

HAT-P-3b

R-1455

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-3_140611 · created 2026-08-01T10:05:05+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2456819.8273 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.121 +/- 0.015
Transit depth (Rp/Rs)^2	1.47 +/- 0.38 [%]
Orbital Inclination (inc)	89.16 +/- 0.83
Airmass coefficient 1 (a1)	0.972 +/- 0.019
Airmass coefficient 2 (a2)	0.0174 +/- 0.0045
Scatter in the residuals of the lightcurve fit is	1.93 %
Best Comparison Star	#3 - [251, 423]
Optimal Method	PSF photometry
Transit Duration (day)	0.098 +/- 0.0024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.121 ± 0.015 vs 0.1063 (deviation 0.67σ)
Duration fit vs published	0.098 d vs 0.0867 d (deviation 3.24σ)
Mid-time O-C	4.0 min
Depth SNR	5.5
Residual scatter	1.93 %

Light curve

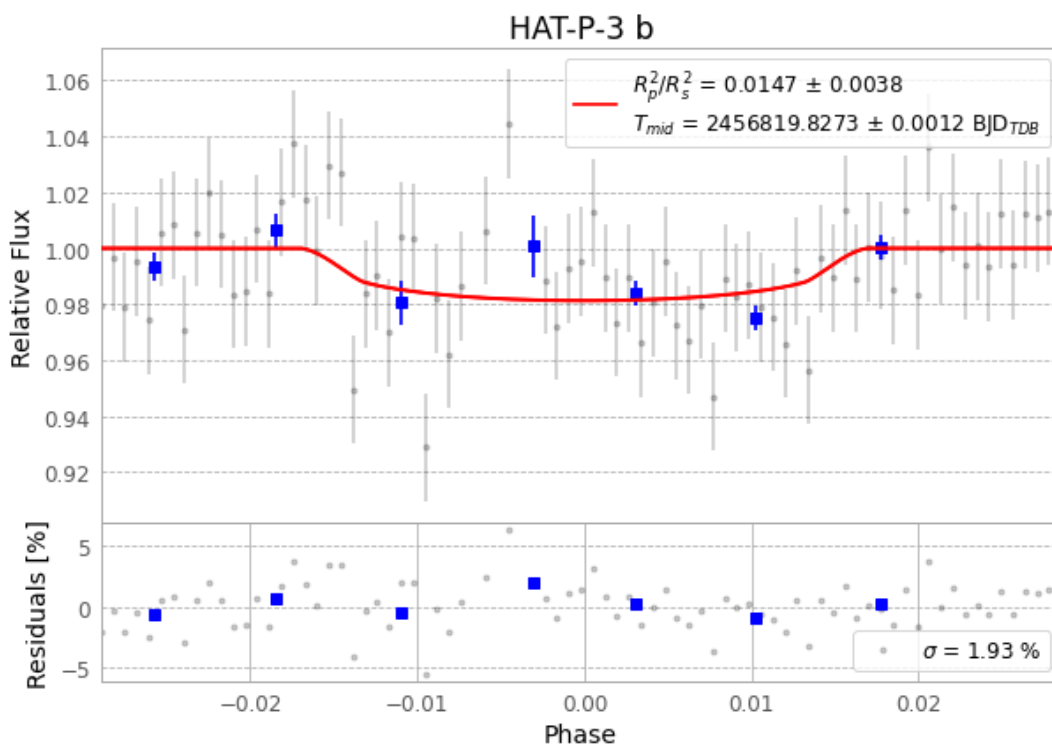


Figure 1 — FinalLightCurve_HAT-P-3 b_2014-06-10.png (SHA-256 e0a19ea024a042b2...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [329, 267], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e33766eba99f7beb...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

80 FITS frames (53 MB), HATP-3140611054812.FITS → HATP-3140611094812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-3-140611.log (aa1926ddeb4e...), FinalLightCurve_HAT-P-3 b_2014-06-10.png (e0a19ea024a0...), FinalLightCurve_HAT-P-3 b_2014-06-10.csv (93d268b94ff7...)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 10:05Z.